

CS-2005 – Database Systems – Spring 2026

Assignment # 01

Maximum Marks: 10

Post Date: February 08, 2026

Due Date: February 20, 2026

Note:

- Plagiarism of any kind will not be tolerated.
- Late submissions will not be accepted.

Question 01

CyberNova Solutions is an international IT services company that provides network security and cloud monitoring solutions to corporate clients. The company operates through multiple branch offices that currently maintain separate file-based systems for storing customer information, incident logs, and service records. Due to the lack of integration between these systems, CyberNova often experiences data duplication, inconsistent records, and slow access to critical information. To overcome these issues, the company plans to develop a centralized, server-based mobile application supported by a Database Management System (DBMS).

1. What limitations of the existing file-based system are faced by CyberNova Solutions in managing its data?
2. How would the use of a Database Management System (DBMS) help CyberNova Solutions manage and store data more efficiently?
3. What role would a server-based mobile application play in improving data accessibility for employees across different branch offices?

Question 02

Given below is a database schema for a cybersecurity services management system that stores information about clients, services, service records, and regional offices.

- **Clients** (client_id, client_name, email, country)
- **Services** (service_id, service_name, service_cost)
- **Service_Records** (record_id, client_id, service_id, service_date, status)
- **Offices** (office_id, office_name, location)

Based on the above schema, write SQL queries for the following:

1. Display the details of all clients belonging to Pakistan.
2. Retrieve the names and costs of services whose service cost is greater than 500.
3. List all service records where the status is 'Completed'.
4. Display the names of clients who belong to Pakistan or the United States.
5. Retrieve service details where the service cost is between 300 and 1000.
6. Display the distinct locations of all offices.
7. Show all service records ordered by service date in descending order.
8. Display client names in uppercase using an appropriate scalar function.
9. Find the total number of service records in the system.
10. Display service names and costs ordered by service cost in ascending order.

Question 03

You are provided with the following snapshot of a relational database that models a library system. The database tracks books, members, borrow records, and librarians. The relevant attributes along with constraints are shown below.

Table 01: Books

Book_ID	Title	Author	Copies_Available
1	"Python Basics"	Ghulam Qadir	3
2	"Cybersecurity 101"	Talha Shahid	2
3	"Database Design"	Abdullah Shaikh	0

Table 02: Members

Member_ID	Member_Name	Email	Membership_Type
101	Aisha	aisha@email.com	Premium
102	Bilal	bilal@email.com	Standard
103	Hina	hina@email.com	Premium

Table 03: Borrow_Records

Record_ID	Member_ID	Book_ID	Borrow_Date	Return_Date
1001	101	1	2026-01-10	2026-01-20
1002	102	2	2026-01-12	NULL
1003	103	3	2026-01-15	2026-01-25

Table 04: Librarians

Librarian_ID	Librarian_Name	Email
201	Nida	nida@email.com
202	Imran	imran@email.com

Constraints

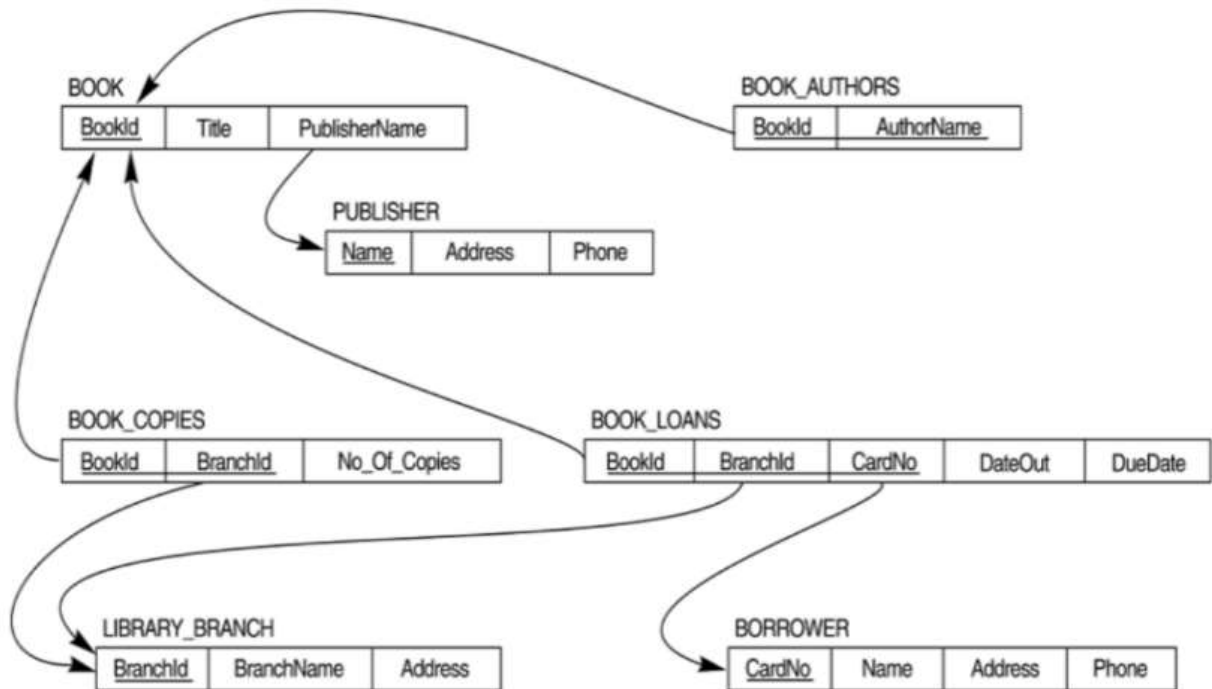
- **Primary Keys:** Book_ID, Member_ID, Record_ID, Librarian_ID
- **Foreign Keys:**
 - Borrow_Records(Member_ID) references Members(Member_ID)
 - Borrow_Records(Book_ID) references Books(Book_ID)
- **Check Constraints:** Copies_Available ≥ 0
- **NOT NULL:** Member_Name, Book_ID, Member_ID in Borrow_Records

Write SQL queries for the following scenarios:

1. Insert a borrow record for a member with Member_ID 104 borrowing Book_ID 1 on 2026-02-05.
2. Update the number of copies available for Book_ID 2 to -1.
3. Delete the member with Member_ID 101 from the Members table.
4. Insert a new member with NULL Member_Name, email 'faizan@email.com', and membership type 'Premium'.
5. Update the borrow record with Record_ID 1002 to reference a non-existent Book_ID 5.
6. Insert a new book with Book_ID 4, title 'Network Security', author 'Ali Raza', and 2 copies available.
7. Update the borrow record with Record_ID 1003 to assign it to Member_ID 102.
8. Delete the book with Book_ID 3 from the Books table.
9. Insert a borrow record with Record_ID 1005, Member_ID 102, Book_ID 3, borrow date '2026-02-07', return date '2026-02-15'.
10. Add a check constraint to the Books table to ensure Copies_Available ≥ 1 .

Question 04

Consider the schema given below and answer the following parts.



1. How many entities (tables) are there in the schema? List their names.
2. List the primary keys for all the tables.
3. Identify which foreign keys exist and explain their purpose.
4. What should be the domain of each attribute for each table?
5. Describe the relationship between Book and Book_Author. How would you represent an author who has written multiple books in this schema?

Question 05

Answer the following questions:

1. What is data independence in a DBMS? Give one simple example showing how changing the storage structure of data does not affect the application program.
2. Give one real-life example where a file-based system is better than a DBMS. Explain in simple terms why a file system might be preferred in that case.
3. What is the difference between DELETE, TRUNCATE, and DROP in SQL? Explain their purpose, use, and effects on the table and data.
4. Why is referential integrity crucial in relational databases, and what problems could arise if it is not enforced?
5. Can a foreign key in a relational database have a NULL value? Explain under what circumstances this is allowed and how it affects referential integrity.

THE END!